A Video-Based Object Tracking Framework using ViBe and Kalman Filter

Project Summary

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1. Introduction

This project is based on a master's thesis completed in January 2019. The implementation and documentation have been reorganized and updated for GitHub publication in 2026.

This document provides a concise overview of the master's thesis project, including the research objective, proposed method, experimental setup, experimental results, main contributions, and future research directions.

Information obtained from images and videos plays an important role in the detection and analysis of moving objects. Computer vision enables a computer to perceive, process visual information, make decisions, and send the required commands to control systems.

Object detection and tracking have a wide range of applications in various fields, including industry, medicine, military systems, security systems, video surveillance, and many other areas.

The presence of a large number of objects, complex backgrounds, camera viewpoints, lighting conditions, and camera characteristics can create challenges in the process of object detection and tracking.

2. Research Objective

The objective of this thesis is to optimize the classical ViBe algorithm in terms of processing time and overall performance. The proposed algorithm was also implemented to evaluate its performance and compare the obtained results with those of the classical algorithm.

Since the classical ViBe algorithm performs poorly in the presence of partial or complete occlusion, the Kalman filter was combined with the classical algorithm to overcome this limitation. The results showed that the proposed algorithm achieved better performance, especially during occlusion, while also providing higher processing efficiency than the classical algorithm.

3. Proposed Method

After receiving the input video, the algorithm converts the video into individual frames and extracts the background model. By comparing the background model with each new input frame and applying a threshold value, the moving object is detected. Then, a morphological filter is used to remove noise and shadows caused by false detections. Next, a Kalman filter is employed to overcome the limitations of the classical algorithm (such as occlusion) and maintain tracking stability. Finally, the background model is updated, and the output image is generated.

In this study, by combining the classical ViBe algorithm with the Kalman filter, we aimed to improve the algorithm in terms of both foreground detection and object tracking. The classical ViBe algorithm provides accurate target detection through precise background modeling. The Kalman filter predicts the target position in subsequent frames and improves tracking stability. As a result, even under challenging conditions that may cause the target to be lost, the proposed algorithm is able to track the target more accurately.

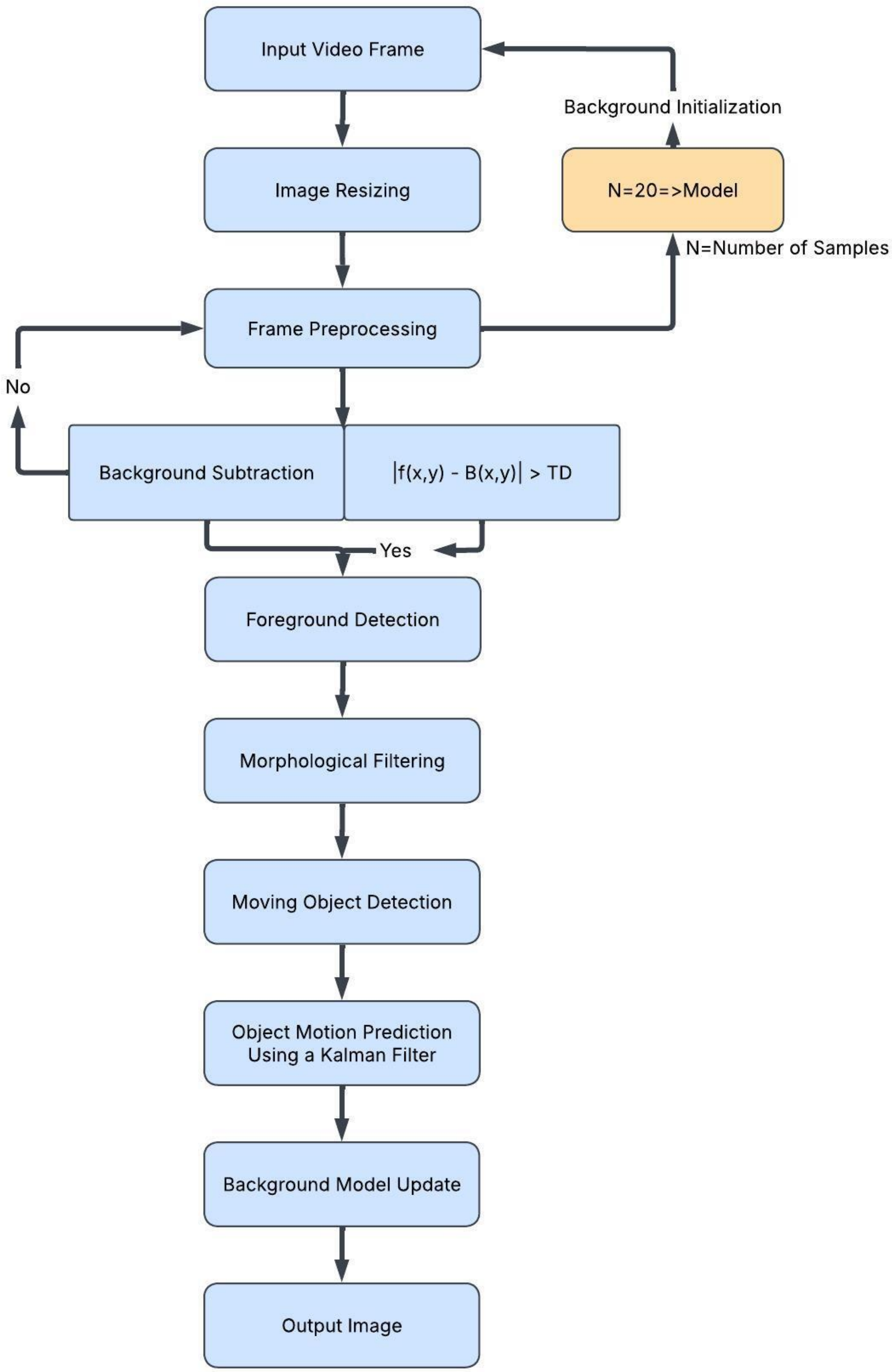

Figure 1. Flowchart of the proposed algorithm.

4. Experimental Setup

The algorithm was implemented in MATLAB. The experimental evaluation was performed using three public video sequences: Box, car-overhead-1, and car-overhead-2-hires. To maintain consistency throughout the implementation and presentation of the results, car-overhead-1 and car-overhead-2-hires are referred to as Car1 and Car2, in the remainder of this report.

The available Ground Truth data were used for each video sequence. The Ground Truth contained the bounding box coordinates for every frame and was used as the reference for performance evaluation.

The algorithm performance was evaluated by comparing the estimated object center with the corresponding coordinates provided in the Ground Truth. The Euclidean distance between these two coordinates was calculated for each frame to measure localization accuracy.

In addition, the estimated x and y coordinates were compared with the corresponding Ground Truth values to analyze the localization error.

5. Results and Discussion

The performance of the proposed algorithm was evaluated against the Mean Shift algorithm using the Euclidean distance. The results showed that the proposed algorithm achieved better performance in some challenging situations (such as the Box video), whereas its performance was comparable to that of the Mean Shift algorithm in the remaining cases. Furthermore, the x and y error plots demonstrated that the proposed algorithm estimated the object position with good accuracy.

The use of the Kalman filter improved tracking stability. In addition, the combination of the Kalman filter and the background subtraction approach improved both localization accuracy and processing speed, resulting in more reliable performance under challenging conditions.

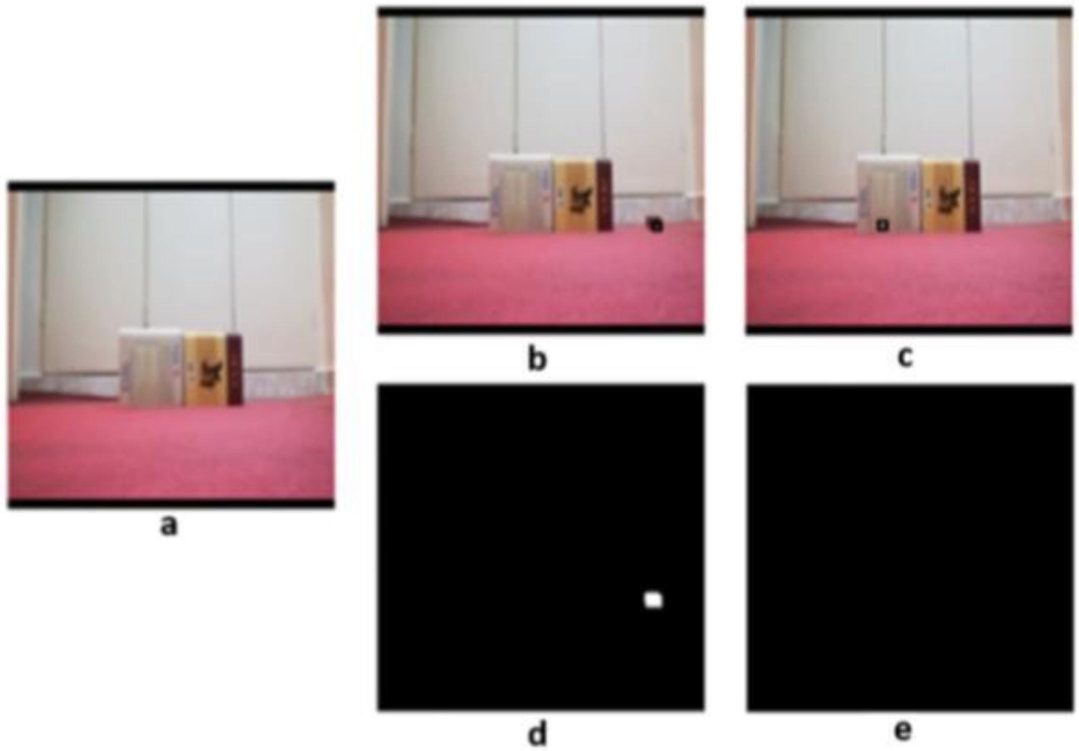

Figure 2. Results of the proposed algorithm on the **Box** video. (a) Background image without the moving object, (b) moving foreground object, (c) occlusion scenario with the object position estimated by the Kalman filter, (d) ground-truth tracking annotation corresponding to (b), and (e) ground-truth tracking annotation corresponding to the occlusion scenario in (c).

6. Contributions

In this study, the classical ViBe algorithm was combined with the Kalman filter to improve moving object localization and tracking. Furthermore, the performance of the proposed algorithm was evaluated using Ground Truth data and the Euclidean distance. In addition, the processing time of the proposed algorithm was compared with those of the classical ViBe and Mean Shift algorithms.

7. Future Work

For future research, the algorithm can be made more robust against noise by applying appropriate techniques and extended for use with videos captured by moving cameras. Intelligent adjustment of parameters and thresholds can also improve the algorithm performance. The algorithm can also be extended to detect multiple moving objects in crowded scenes with multiple targets. Furthermore, improving the algorithm performance under scale variations will also be an appropriate topic for future studies.